

UPDATE:



Appendix C

Federal Performance Measures

APPENDIX C — Federal Performance Measures

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Background

Federal legislation, starting with MAP-21 and continuing in the subsequent FAST Act and IIJA, requires state departments of transportation (DOTs) and metropolitan planning organizations (MPOs) to establish and use a performance-based approach in transportation decision-making to achieve national goals. Performance-Based Planning and Programming (PBPP) is a systematic, data-driven approach to transportation decision-making, ensuring that resources are allocated efficiently to achieve the identified performance outcomes.

The goal of PBPP is to ensure targeted investment of federal transportation funds by increasing accountability and transparency and providing better investment decisions that focus on key outcomes related to the national performance goals:

1. Safety
2. Infrastructure preservation
3. Congestion reduction
4. System reliability
5. Freight movement and economic vitality
6. Environmental sustainability
7. Reduced project delivery delays

State DOTs, MPOs, and transit providers identify and track the desired performance outcomes through Transportation Performance Management (TPM). TPM includes tracking performance measures, setting data-driven targets for each

measure, and selecting projects to help meet those targets. The FAST Act and IIJA also require that the TIP include a description of its anticipated effect toward achieving the established performance targets, linking investment priorities to those performance targets.

The FHWA has established three performance measures for roadway performance. The three performance measures are:

1. Roadway Safety (Performance Measure Rule 1 [PM1]);
2. Bridge and Pavement Condition (Performance Measure Rule 2 [PM2]); and
3. System Performance (Performance Measure Rule 3 [PM3]), which evaluates system reliability, freight reliability, and the federal Congestion Mitigation and Air Quality (CMAQ) program.

State DOTs are required to establish targets for each performance measure and report progress toward these targets, except for the CMAQ program Urban Area (UA) measures, where DOTs and MPOs collaborate to develop the unified target. MPOs, such as DVRPC, must either support the respective state DOT and transit operator targets or create their own regional targets. As a bi-state MPO, DVRPC must plan and program projects to contribute toward separate sets of targets: one set for each state in which the Planning Area boundary extends. DVRPC has agreed to support the PM2 and PM3 targets set by the Pennsylvania Department of Transportation (PennDOT) and the New Jersey Department of Transportation (NJDOT), respectively. Written procedures were developed between the

state DOTs and MPOs regarding the coordination of TPM activities.

For PM1, Roadway Safety, DVRPC has established its own target in alignment with DVRPC's Regional Vision Zero 2050 goal. If an MPO adopts regional targets, it must adopt targets that cover the entire MPO region.

There are multiple performance measures established within the three performance measure groupings. Each performance measure grouping has different requirements for reporting and updating performance targets. Table C-1 summarizes these measures, the area for which they are being reported, the facilities included, and the update frequency.

On the transit side, the Federal Transit Administration (FTA) has established performance measures for Transit Asset

Management (TAM) and Transit Safety. Table C-2 has a summary of these measures. FTA regulations establish a strategic and systematic process of operating, maintaining, and improving public capital assets effectively through their life cycle. The performance management requirements are a minimum standard for transit operators.

DVRPC has agreed to support the targets for transit assets and transit safety set by the region's transit operators: the Southeastern Pennsylvania Transportation Authority (SEPTA), NJ Transit, and DRPA/PATCO. Additionally, DVRPC has a memorandum of agreement with the respective transit agencies that outlines the requirements and responsibilities under the TPM process. Transit agencies are required to submit their performance targets to the National Transit Database (NTD) annually, as well as a supporting narrative that reports progress towards these targets.

Table C-1: FHWA Performance Measures Summary

Goal Area	Performance Measure	Geography	Network	Reporting Frequency
PM1 Roadway Safety	Number of Fatalities	Statewide or Regional	All Roads	Annual
	Fatality Rate (per 100 million VMT)			
	Number of Suspected Serious Injuries			
	Suspected Serious Injury Rate (per 100 million VMT)			
	Number of Non-Motorized Fatalities and Suspected Serious Injuries			
PM2 Bridge and Pavement Condition	Good Pavement Lane Miles	Statewide or Regional	Interstates and National Highway System (NHS)	Two-Year Interim Target, Four-Year Target
	Poor Pavement Lane Miles		NHS	
	Good Bridge Deck Area			
	Poor Bridge Deck Area			
PM3 System Performance	Person Miles Traveled with Reliable Travel Times (%)	Statewide or Regional	Interstates and NHS	Two-Year Interim Target, Four-Year Target
	Truck Travel Time Reliability Index		Interstates	
	Percentage Non-SOV Travel	Urban Area (UA) with a population over 200,000	All UAs	
	Annual Hours of Peak Hour Excessive Delay (PHED) per Capita		All NHS roads within UAs, AM and PM Peak Periods	
	CMAQ pollution reduction outcomes	Statewide and Regional	CMAQ Projects	

Source: DVRPC, 2025.

Table C-2: FTA Performance Measures Summary

Goal Area	Performance Measure	Assets/Geography	Reporting Frequency
Transit Assets	Percentage of Rolling Stock Meeting or Exceeding the Useful Life Benchmark	Revenue Vehicles	Annual
	Percentage of Support Equipment Meeting or Exceeding the Useful Life Benchmark	Non-Revenue Vehicles	
	Percent of Facilities Rated Below 3 on the TERM Scale	Passenger, Administrative, and Maintenance Facilities	
	Percent of Track Segments with Performance Restrictions	Rail Track	
Transit Safety	Fatalities (Total and Number per Vehicle Revenue Mile)	Entire Transit Agency Service Area	Annual
	Injuries (Total and Number per Vehicle Revenue Mile)		
	Safety Events (Total and Number per Vehicle Revenue Mile)		
	System Reliability (Mean Distance in Miles – Vehicle Revenue Miles – Between Major Service Failures)		
	Assaults on Transit Workers (Total and Number per Vehicle Revenue Mile)		

Source: DVRPC, 2025.

Historical data for the metrics tracked through TPM are available on DVRPC's [Tracking Progress](#) website.²⁴ *Tracking Progress* is an interactive dashboard for exploring regularly updated data to gauge our progress toward achieving the vision established in DVRPC's Long-Range Plan. Each indicator includes a time series analysis to track data trends at different scales, including the regional level, county level, and by UA or transit agency, depending on the metric. The dashboard tracks actual performance for each of the performance measures via the following metrics:

- Transportation Safety
- Bridge Conditions
- Pavement Conditions
- Congestion
- Transit Conditions

The following sections of this Appendix provide more information on each performance measure and its associated targets, as they pertain to the region. For more information about the development and implementation of TPM policy and rulemaking, see www.fhwa.dot.gov/tpm. To view the latest TPM targets and performance for DVRPC, visit www.dvrpc.org/tpm.

²⁴ *Tracking Progress*, Delaware Valley Regional Planning Commission, accessed June 8, 2025, www.dvrpc.org/trackingprogress/

Roadway Performance Measures

PM1: Roadway Safety

Targets and Reporting Requirements

PM1 Roadway Safety includes five performance measures:

- Number of Fatalities
- Annual Fatality Rate, per 100 million Vehicle Miles Traveled (VMT)
- Number of Suspected Serious Injuries
- Suspected Serious Injury Rate, per 100 million VMT
- Number of Non-Motorized Fatalities and Suspected Serious Injuries

Targets are updated annually and measure performance for all roads in a region or state. Targets are reported using five-year rolling averages, with baseline data projected to a two-year target. Current targets and performance data can be found on DVRPC's TPM website under the [Roadway Safety](#) tab.²⁵

If an MPO adopts regional targets, it must submit a progress report to the respective state DOT. FHWA will determine whether a state has met or made significant progress toward its safety performance targets. A state is considered to have met or made

²⁵ Delaware Valley Regional Planning Commission. *Transportation Performance Management: Highway*. www.dvrpc.org/tpm/?indicator=highway.

significant progress when at least four out of the five safety performance targets are met, or the actual outcome for the safety performance target is better than baseline performance. New Jersey and Pennsylvania have not met or made significant progress towards their previous roadway safety targets. The penalty for not meeting targets or making significant progress is that the state DOT must:

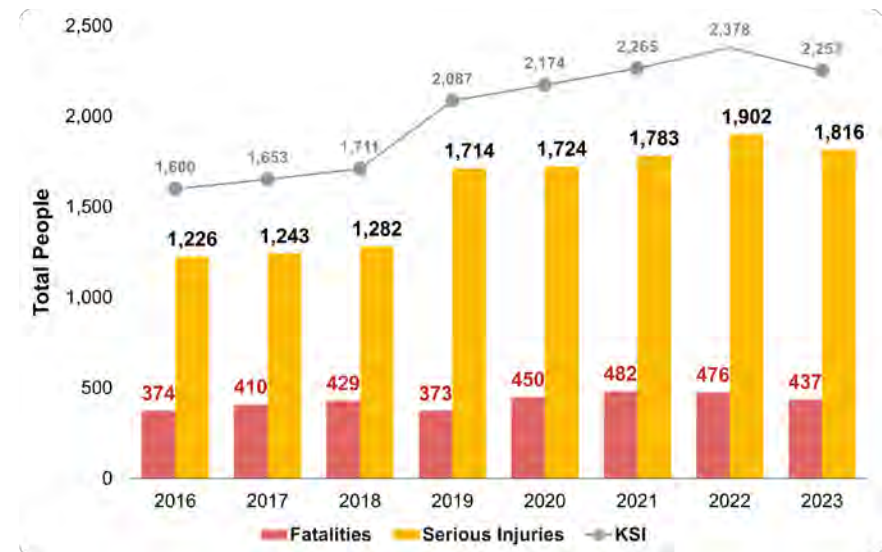
- Submit a Highway Safety Improvement Program (HSIP) Implementation Plan.
- Use obligation authority equal to the HSIP apportionment for the prior year, only for highway safety projects.

Trends

The metric most used to measure roadway safety is “people killed or suspected of being seriously injured” (KSI), which refers to the total number of individuals who either died or sustained severe injuries in a traffic collision. KSI is promoted by FHWA and utilized by both NJDOT and PennDOT.

Regionally, the total KSI on roadways has been increasing overall in the past decade. In 2023, KSI totaled 2,253 people in the region. This was lower than the region’s peak of 2,378 in 2022, but notably higher than the 2010s, when this number was consistently below 2,000. The region has experienced a similar upward trend in both KSI per 100,000 people and KSI per 100 million VMT, with a significant portion of this increase attributable to the 2019 change in the definition for suspected serious injuries.

Figure C-1: KSI on Roadways in the DVRPC Region



Source: Pennsylvania Department of Transportation (PennDOT) Crash Information Tool; New Jersey DOT Crash Records, 2025DVRPC and State DOT Efforts Towards PM1 Targets

Update: Connections 2050 aims to direct 5.9 percent of total roadway revenues in Pennsylvania and 2.0 percent in New Jersey toward substantive safety improvements—projects that go beyond existing design standards to address locations with a high number of crashes and advance Vision Zero goals. While these targeted investments are a smaller share of total roadway funding, they are designed to deliver outsized safety benefits. Most reasonably anticipated roadway funds will be directed toward system preservation projects (70.1 percent in Pennsylvania and 54 percent in New Jersey), which will also enhance safety by modernizing infrastructure to meet current

design standards and improve overall roadway condition. For a list of TIP-funded projects related to Roadway Safety, see chapter four of the New Jersey and Pennsylvania TIPs.

DVRPC has incorporated the Regional Vision Zero policy into the work of the Regional Safety Task Force (RSTF), reframed to embrace FHWA's safe system approach.²⁶ The FHWA Safe System approach is a comprehensive, proactive strategy designed to eliminate fatal and serious injuries for all road users by acknowledging human vulnerability, designing forgiving infrastructure, and ensuring shared responsibility across five key areas: users, vehicles, speeds, roads, and post-crash care. These holistic changes help to advance our safety culture and increase the priority of safety initiatives. In 2023, DVRPC launched the *Regional Vision Zero 2050* program initiative, funded by a Safe Streets and Roads for All (SS4A) grant. This effort involves close coordination with county-partner sub-awardees to develop a plan that facilitates collaborative engagement with municipal partners. DVRPC continues to include crash analyses in our work program projects to advance substantive infrastructure safety improvements. The DVRPC region has received numerous SS4A grants over the short life of this program. This includes two grants for DVRPC, as well as grants for several counties and many municipalities.²⁷

DVRPC has developed a Regional High Injury Network (RHIN), the result of a data-driven process designed to show where KSI, bicyclist, and pedestrian crashes occurred using reportable

crash data. DVRPC published the RHIN in advance of the *Regional Vision Zero Safety Action Plan* release to support local safety planning.

The *Transportation Safety Analysis and Plan* (TSAP) is the regional safety planning document informing safety work across the Greater Philadelphia area. Close collaboration with partners helps to raise the profile of regional safety needs and connects them to funding opportunities.

In Pennsylvania, DVRPC has developed a road diet network screening analysis for PennDOT District 6-0, and staff continues to screen roadway maintenance plans for Complete Streets facility opportunities as part of the PennDOT Complete Streets Resurfacing program (formerly the Bicycle Friendly Resurfacing program). DVRPC has conducted two High Injury Network (HIN) safety corridor studies in the City of Philadelphia, which have received federal implementation funding. One additional corridor safety study is ongoing. Additionally, DVRPC explored the possibility of a context-based speed limit setting to address speeding-related crashes on urban arterials in partnership with the City of Philadelphia.

The City of Philadelphia regularly releases a five-year *Vision Zero Action Plan*. The *Action Plan* guides Philadelphia's traffic safety initiatives and is accompanied by a capital plan, which focuses on potential safety solutions for priority corridors and intersections on the City's HIN.

²⁶ "Zero Deaths and Safe System | FHWA." n.d. Highways.dot.gov. <https://highways.dot.gov/safety/zero-deaths>.

²⁷ For a list of competitive grant awards in the DVRPC region, see www.dvrpc.org/pdf/summaryofijjaawardstothedvrpc%20region.pdf.

In New Jersey, DVRPC-led safety planning efforts include Road Safety Audits (RSAs) covering all four counties, funded by NJDOT. The completed RSAs are now advancing into the HSIP Local Safety Program application process. In Trenton, this has included a *Complete Streets Design Handbook*, as well as various published and upcoming plans relating to the bike network, trails planning, and Vision Zero. Other studies include *the Brown Mills Bicycle and Pedestrian Action Plan* and the *Harrison Avenue Pedestrian Road Safety Audit*.

PM2: Roadway Infrastructure Condition

Targets and Reporting Requirements

PM2 Bridge and Pavement Condition measures the performance of:

- Interstate Pavement Infrastructure
- Non-Interstate National Highway System (NHS) Pavement Infrastructure
- NHS Bridge Infrastructure

Performance is measured in terms of the percentage of pavement lane miles or bridge deck area in “good” condition and “poor” condition.

FHWA has set standards that categorize pavement and bridge conditions as good, fair, or poor. Pavement condition is determined by specific measures, which include the International Roughness Index (IRI), cracking, rutting, and faulting. Bridge condition is based on ratings given to the bridge deck, superstructure, and substructure components during regular inspections using National Bridge Inventory (NBI) standards. If any of the components of a structure qualify as poor, the entire bridge structure is deemed poor. All components of the bridge structure must have a good rating for the structure to be deemed in good repair. Bridge condition performance measures are calculated by summing the deck area of bridges in “good” and “poor” condition and dividing by the total deck area of all NHS bridges. Poor condition does not mean the bridge is unsafe to use. Both PennDOT and NJDOT will take necessary action to restrict heavy-weight vehicles or close a bridge to ensure safety.

State DOTs must establish pavement targets, regardless of ownership, for the full extent of the Interstate and non-Interstate NHS.²⁸ State DOTs must establish targets for all bridges carrying the NHS, which includes on- and off-ramps connected to the NHS within a state, and bridges carrying the NHS that cross a state border, regardless of ownership.

DOTs and MPOs must then report on performance and set two- and four-year expected performance targets. Current targets and

²⁸ The NHS was designated in 1995 and expanded in 2012. It contains nearly 221,000 miles of roadway across the U.S., including all interstates and principal arterials, the Strategic Highway Network for defense purposes, and intermodal connectors, which are

roads connecting the NHS to ports, airports, rail terminals, and pipeline terminals. The NHS covers just 5.3 percent of the nation’s roads, but it carries about 55 percent of all traffic.

performance data can be found on DVRPC's TPM website under the Bridge and Pavement Condition tab.²⁹

Federal regulations require DOTs to (1) submit a Transportation Asset Management Plan (TAMP) that, at a minimum, forecasts asset deterioration, determines costs and benefits over an asset's life cycle, and identifies short- and long-term budget needs; and (2) produce a recommended program that is financially constrained.

The federal Infrastructure PM Rule requires that less than 5 percent of Interstate lane miles be in poor condition to meet the federal threshold for pavement condition. If pavement conditions on the Interstate system fail to meet the federal minimum condition thresholds in the most recent year of the State biennial report, the state DOT must comply with the provisions outlined in 23 USC 119(f) for the subsequent fiscal year. The State must:

- Use obligation authority to transfer a portion of State Transportation Planning (STP) funds to the National Highway Performance Program (NHPP) for maintenance projects to address Interstate pavement conditions.
- Submit a TAMP that describes actions the state will take to meet or make significant progress toward meeting its targets. The TAMP should guide the state's project decisions to meet or make substantial progress toward

meeting its infrastructure performance targets in subsequent years.

If a state has not met or made significant progress toward meeting its targets on the non-Interstate NHS system, the state DOT must submit a TAMP that describes actions the state will take to meet or make significant progress toward meeting its targets. The TAMP should guide the state's project decisions to meet or make significant progress toward achieving its infrastructure performance targets in subsequent years.

The federal Infrastructure PM Rule requires that no more than 10 percent of the total deck area of bridges on the NHS be considered structurally deficient to meet the federal threshold for bridge condition. If a state has not met the federal threshold for bridge conditions for three consecutive years, the state DOT must obligate and set aside NHPP funds for eligible NHS bridge projects. The set-aside will remain in effect until the State meets the threshold of less than 10 percent of bridge deck area classified as structurally deficient.

Trends

In 2023, 4.2 percent of the region's NHS pavement lane miles were in poor condition, according to the PM2 metric, which is one percentage point lower than in 2020. Just 0.5 percent of the region's Interstates were in poor condition in 2023, down slightly from 1 percent in 2020. Five percent of non-Interstate NHS

²⁹ Delaware Valley Regional Planning Commission. *Transportation Performance Management: Bridge & Pavement Condition*. www.dvrpc.org/tpm/?indicator=bridgepavement.

pavement was in poor condition in 2023, down slightly from 6 percent in 2020.

Over the last two decades, the region has made significant progress in improving NHS bridge conditions. Since peaking at nearly 19 percent in 2004, deficient NHS deck area has decreased to just below 6 percent in 2024—a 69 percent drop in the total deficient NHS deck area. The total number and deck area of deficient NHS bridges have decreased steadily.

DVRPC and State DOT Efforts Towards PM2 Targets

The Greater Philadelphia region remains committed to preserving and maintaining pavement and bridge infrastructure. The current TIP and Plan continue to emphasize analysis related to transportation system preservation needs and funding, aligning with the support of pavement and bridge condition performance targets. This, in turn, informs the fiscally constrained list of projects included in the Plan and TIP. For a list of TIP-funded projects related to Bridge and Pavement Condition, see chapter four of the New Jersey and Pennsylvania TIPs.

NJDOT's TAMP reviews the current state of physical assets; identifies objectives for asset condition and performance; analyzes future conditions under different investment levels; determines the best investment strategies for assets across their life cycle, given the investment level; and considers risks to accomplishing the objectives and implementing planned investment strategies.

PennDOT's pavement condition targets are consistent with its asset management objectives to maintain the system at the desired state of good repair (SGR), manage for the Lowest Life-Cycle Cost (LLCC), and achieve national and state transportation goals. LLCC is a tool to determine the best option by considering all transportation agency expenditures and user costs throughout the life of an alternative, not just the initial investment. PennDOT's RoadCare pavement program can project pavement conditions over time based on investment levels, and identify when preservation projects are needed and estimate their associated costs. Likewise, PennDOT's BridgeCare software can project future bridge projects and conditions at various investment levels. Both programs utilize an LLCC approach. PennDOT is working on a companion program, Project Builder, which will identify corridors in need of preservation, rather than focusing on individual facilities, pavement, and bridges in silos.

PM3: System Performance

Targets and Reporting Requirements

PM3 measures roadway system reliability, freight reliability, and performance of the Congestion Mitigation and Air Quality (CMAQ) program. Current targets and performance data can be

found on DVRPC's TPM website under the System Performance tab.³⁰ The measures and metrics include:

- **Roadway System Reliability:** Includes two reliability measures, one each covering the Interstate System and the non-Interstate National Highway System (NHS). Reliability is measured as Level of Travel Time Reliability (LOTTR), which indicates the percentage of person miles traveled that are reliable on the Interstate and non-Interstate systems within a region. Travel times, VMT, and average vehicle occupancy are factored into this reliability measure to calculate the percentage. LOTTR is measured during four peak travel periods, which include peak daytime periods and weekend periods. LOTTR indicates roadway travel time reliability measured by the ratio of the 80th percentile travel time to a "normal" travel time (50th percentile). Any roadway segment with an LOTTR value of 1.50 or more is considered unreliable (see Figure C-2).
- **Freight Reliability:** Freight reliability is measured for the Interstate system only, using a Truck Travel Time Reliability (TTTR) metric. State DOTs must establish a single index for the Interstate system in the state for five travel periods, which include morning and evening weekday peaks, a mid-day, an overnight, and weekend periods. The TTTR indicates the reliability of the Interstates for freight movement, measured by the ratio of the 95th percentile travel time to a "normal" travel time (50th percentile). Unlike LOTTR, there is no threshold

established for unreliability; the higher the index, the more unreliable (see Figure C-3).

- **Congestion Mitigation and Air Quality (CMAQ)**

Performance: There are three performance measures established to measure the performance of the federal CMAQ program. Two of the measures assess the program's influence on traffic congestion, and the third measures the program's performance in reducing on-road mobile source air pollution.

- o **Congestion Measures:** As part of the CMAQ Program, congestion measures only apply on the NHS in UAs that are in all or part of a designated "nonattainment" or "maintenance area" under the Clean Air Act. Roadways outside the UAs are excluded from the measures. The measures also only apply to UAs that contain populations over 200,000. Most of the DVRPC region is in the Philadelphia UA, but a portion of the DVRPC region also encompasses parts of New York City, Allentown-Bethlehem-Easton, and Trenton UAs. There are two performance measures under this category:
 - Annual Peak Hour Excessive Delay (PHED) per Capita helps to identify areas with excessive traffic congestion and assess pollutants in a region.

³⁰ Delaware Valley Regional Planning Commission. *Transportation Performance Management: System Performance*. www.dvrpc.org/tpm/?indicator=systemperf

- Percent Non-Single Occupant Vehicle (non-SOV) Travel. Reducing this percentage may have beneficial effects for reduced congestion, improved air quality, and improved health.
- *Air Pollution Reduction Measures.* The federal rules requiring the assessment of the CMAQ program's effectiveness in reducing on-road mobile air pollution require that state DOTs and MPOs serving UAs with over 200,000 people

establish two- and four-year targets for the amount of applicable air pollutants that are reduced through CMAQ-funded projects for both the MPO region and statewide. Applicable air pollution is determined by the attainment status of the criteria pollutants. If the states and MPOs do not meet these targets, they are able to adjust the targets and evaluate future CMAQ investments that may improve progress towards meeting the targets.

Figure C-2: LOTTR Interstate and Non-Interstate National Highway System Roadways

Level of Travel Time Reliability (LOTTR)

- Greater than 2.50
- 2.00 to 2.49
- 1.50 to 1.99
- Less than 1.50

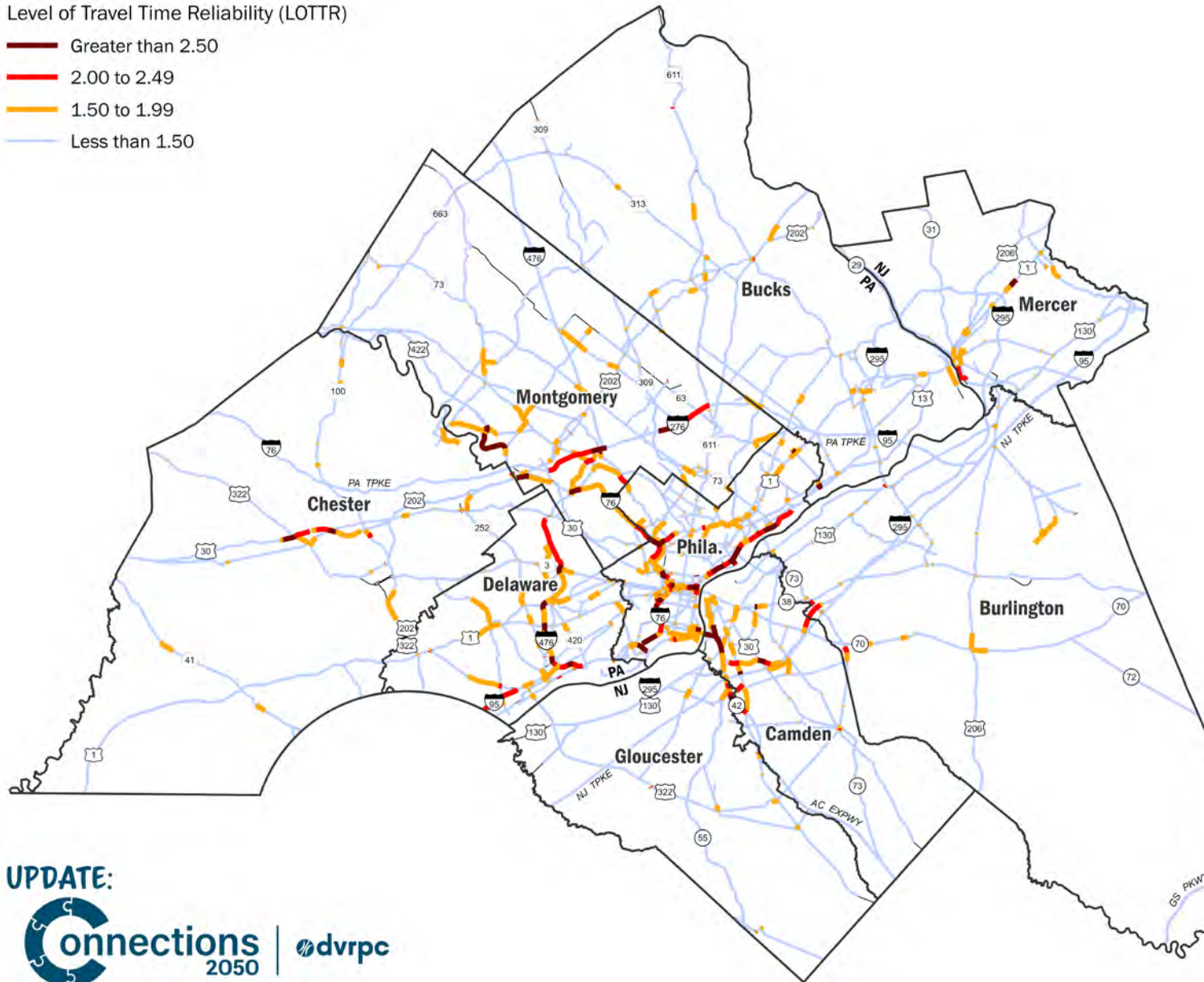


Figure C-3: TTTR Interstate Roadways

Truck Travel Time Reliability (TTTR)

- Greater than 2.50
- 2.00 to 2.49
- 1.50 to 1.99
- Less than 1.50

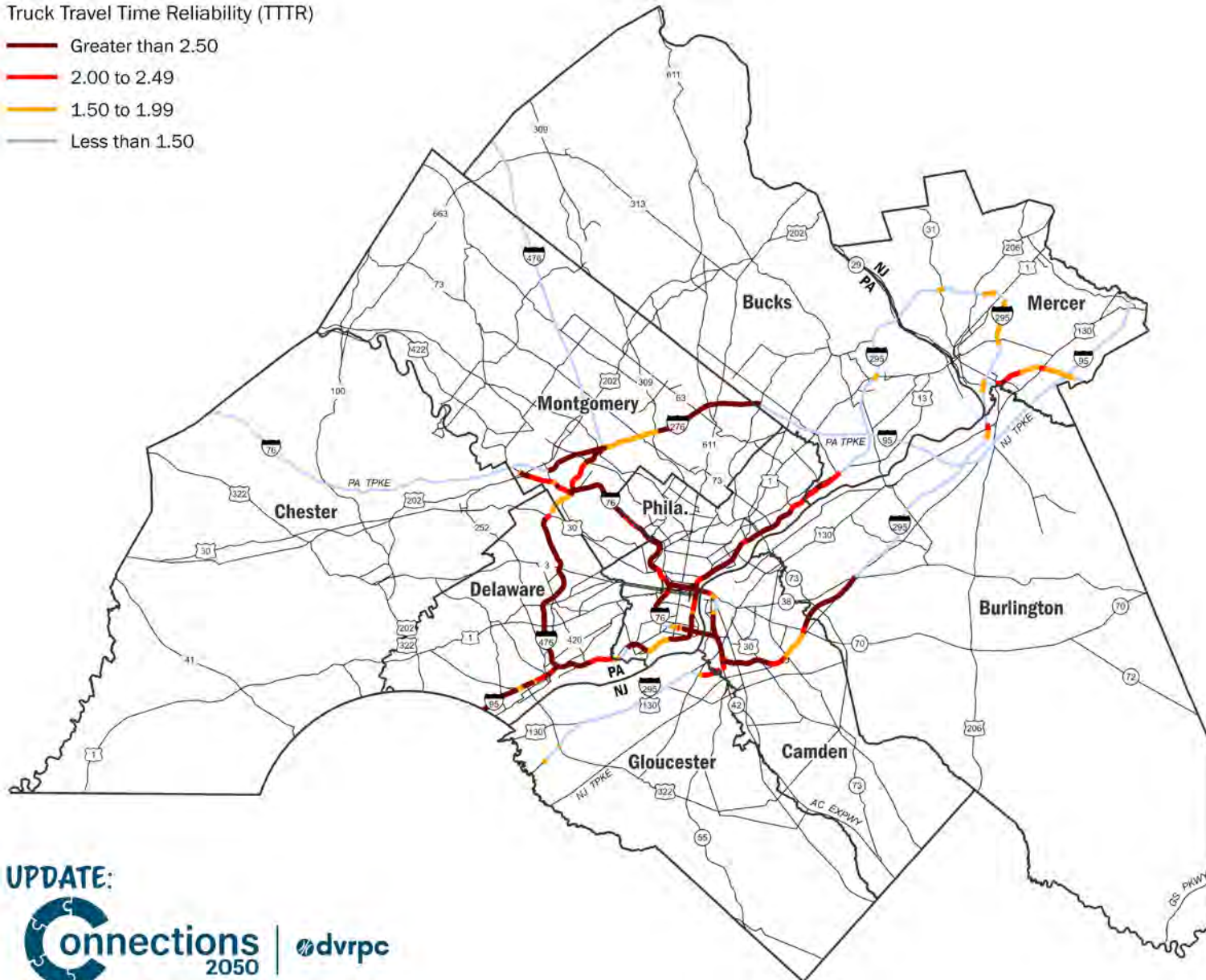
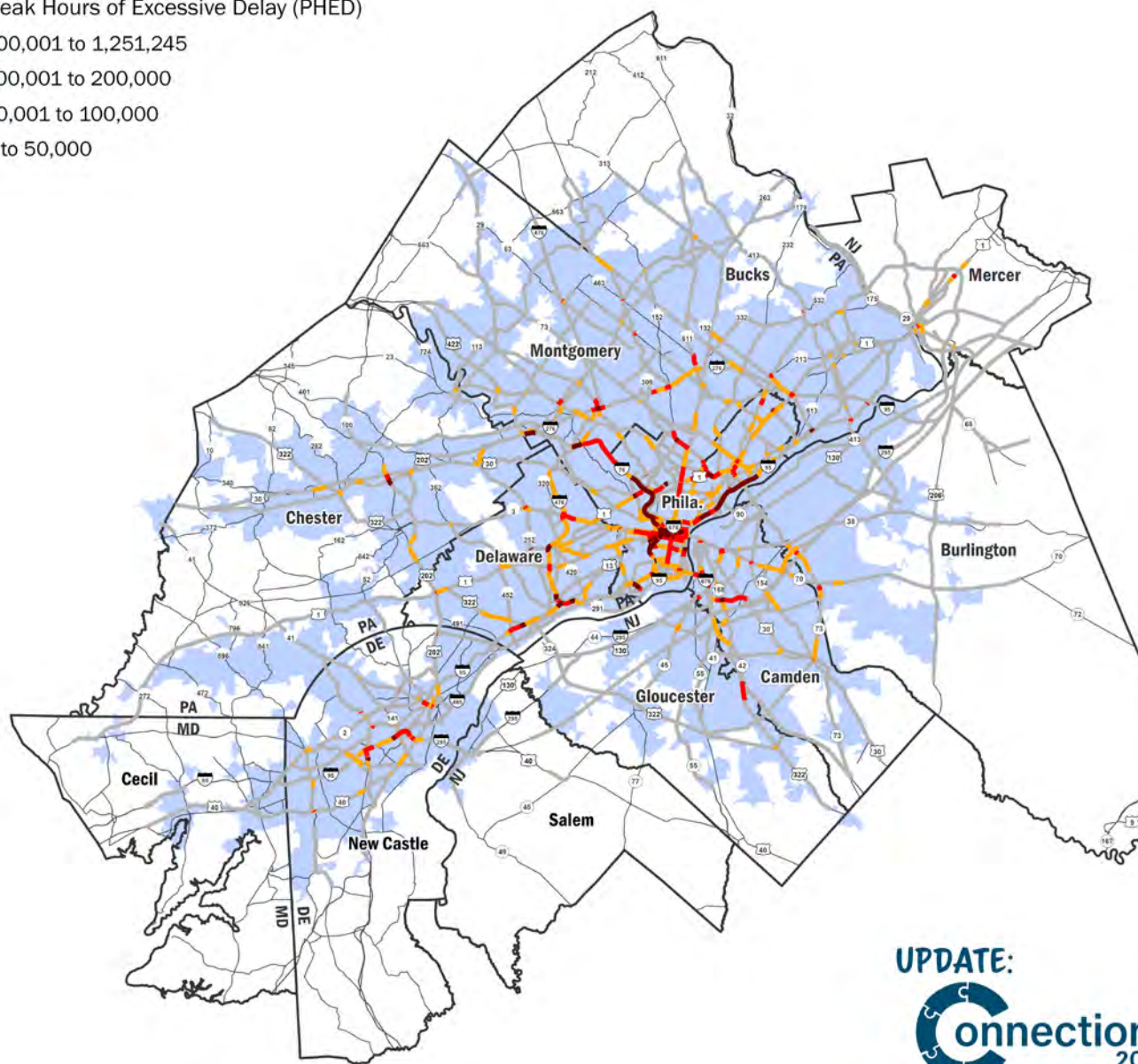


Figure C-4: PHED on the National Highway System in the Philadelphia PA-NJ-DE-MD Urban Area

Annual Peak Hours of Excessive Delay (PHED)

- 200,001 to 1,251,245
- 100,001 to 200,000
- 50,001 to 100,000
- 0 to 50,000



Sources:
DVRPC, 2024;
INRIX, 2023

Trends

Regionally, roads have become more reliable since 2017 on both the Interstate and Non-Interstate NHS. As of 2023, 90.7 percent of Non-Interstate NHS and 81.0 percent of Interstate roadways scored as reliable using the LOTTR metric. This is compared to 84 percent and 74 percent, respectively, in 2017. TTTR has also decreased since 2017. As of 2023, the average Interstate TTTR score was 1.67, down from 1.83 in 2017. PHED has also decreased overall since 2017 for the DVRPC region. In the Philadelphia UA, there were 78.6 million annual hours of PHED in 2023, or 13.9 hours of PHED per capita, compared to 88.1 million hours in 2017. In the Trenton UA, PHED increased from 1.0 million in 2021 to 1.4 million in 2023, or 4.4 hours of PHED per capita. Each of these measures of travel congestion experienced a decrease in delay and an increase in system reliability in 2020, due to the reduction in traffic during the COVID-19 pandemic. Total congestion has returned to pre-COVID levels but is now generally more evenly spread throughout the day than it was before the pandemic.

DVRPC and State DOT Efforts Towards PM3 Targets

DVRPC is committed to enhancing road reliability within the region. Staff collaborate with PennDOT and NJDOT, as well as county, city, and transit partners, to develop projects that enhance travel time reliability and contribute to meeting state targets.

DVRPC proactively seeks to include freight as a primary planning factor in its Long-Range Plan and TIP development, as well as through the conduct of technical studies. At the forefront of DVRPC's Freight Planning Program is the Delaware Valley Goods Movement Task Force, a broad-based freight advisory committee that provides a forum for the private and public sector freight communities to share their unique perspectives on regional plans and specific projects. The FAST Act established, and the IIJA continued, the National Highway Freight Program (NHFP) to improve the efficient movement of freight on the National Highway Freight Network (NHFN). The NHFP's eligibility criteria require projects to contribute to the efficient movement of freight and be identified in the state's freight investment plan. States may use up to 10 percent of NHFP funding each year for public or private freight rail, water facilities (including ports), and/or intermodal facilities. For a list of projects funded with federal NHFP and NHPP funds in the DVRPC region, see chapter four of the New Jersey and Pennsylvania TIPs for more information.

DVRPC's *Congestion Mitigation and Air Quality Interim Performance Plan (2022–23)*³¹ identifies all TIP projects that will help the MPO and states meet the four-year targets for traffic congestion. DVRPC's CMP facilitates a CMP Advisory Committee and generates a list of the top-most congested roadway facilities and bottleneck locations for state, county, and local roadways.

³¹ *Congestion Mitigation and Air Quality Program Interim Performance Plan (2022–2023)*, Delaware Valley Regional Planning Commission, July 2022, www.dvrpc.org/products/25117.

Transit Performance Measures

The FTA has established performance measures for Transit Asset Management and Transit Safety (see Table C-2). FTA regulations establish a strategic and systematic process of operating, maintaining, and improving public capital assets effectively through their life cycle.

Transit Asset Management

Targets and Reporting Requirements

Transit providers are required to coordinate with MPOs to set performance targets for assessing the state of good repair (SGR) of four capital-asset categories: Rolling Stock, Equipment, Facilities, and Infrastructure. Current targets and performance data can be found on DVRPC's TPM website under the [Transit Asset Management](#) tab.

A transit asset is in an SGR if:

1. It can perform its designed function;
2. It does not pose an unacceptable safety risk; and
3. Its life-cycle investments have been met or recovered.

Rolling Stock: Percent of the transit provider's rolling stock that has met or exceeded their Useful Life Benchmark (ULB).

The performance-based planning regulation requires that transit providers set annual targets for the average age of their revenue-

generating vehicles (rolling stock) for each of 23 vehicle types. The targets specify the percentage of the transit provider's rolling stock that exceeds (is older than) the Useful Life Benchmark (ULB) measured in years. The ULB is established by the transit provider with FTA guidance.

Equipment: Percent of the transit provider's non-revenue vehicles that have met or exceeded its ULB.

Transit providers are required to set annual targets for the average age of their non-revenue generating equipment for automobiles, rubber-tired vehicles, and steel-wheeled vehicles. Targets are only set for vehicles that are road-worthy, self-propelled, maintenance, or construction vehicles. The targets are set for the percentage of equipment by vehicle category type that exceeds the non-revenue generating fleet's ULB.

Facilities: Percent of assets with condition rating below 3.0 on the FTA Transit Economic Requirements Model (TERM) Scale for Passenger Facilities and Administration Facilities.

Transit providers are required to set annual targets for the percentage of facilities (by group) that are rated less than 3.0 on the TERM Scale. The two groups of facility types are Passenger and Administration Facilities. FTA requires that facility condition data be fully updated every four years, at a minimum. Agencies may choose to assess a quarter of their facilities every year, or more frequently. Each annual report must include updated facility condition data based on any assessments completed since the last report. Only facilities with direct capital responsibility require condition assessments. For example, a

facility such as Philadelphia's 30th Street Station, which is owned and maintained by Amtrak, is not subject to a local transit agency's capital responsibility and therefore is not included in that agency's performance reporting.

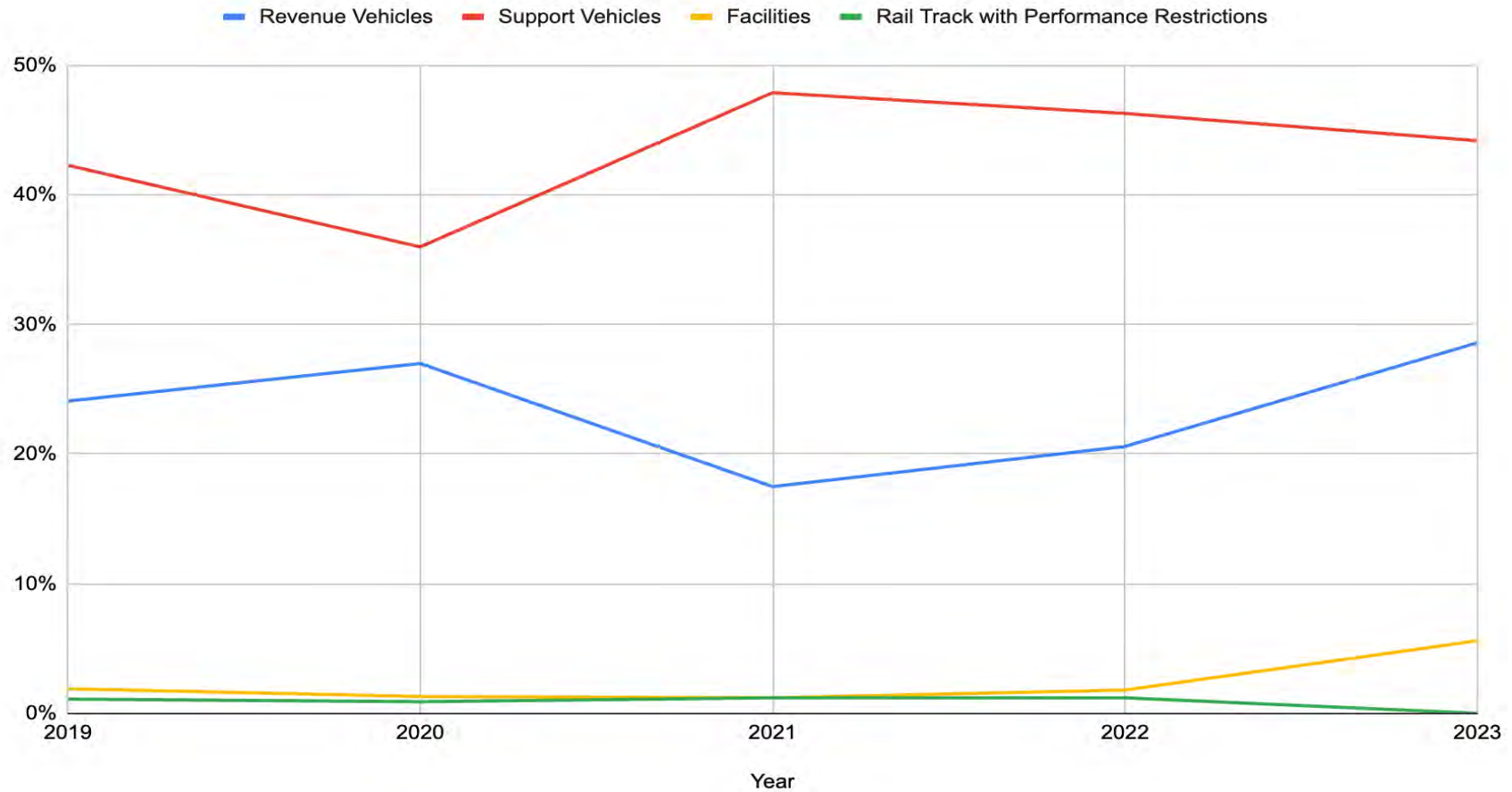
Infrastructure: Percent of the transit provider's fixed guideway track miles that have performance restrictions. The annual performance measure for infrastructure is an average of each month's performance restriction measure.

Transit providers are required to set annual targets for the percentage of fixed guideway track segments that have performance restrictions. A performance restriction is defined as a segment of guideway track miles where the maximum permissible speed of transit vehicles is below the guideway's full-service speed. Restrictions can be caused by issues with rail fixed guideway, track, power, or signal systems. Performance restrictions should be measured at 9:00 AM local time on the first Wednesday of each month. Regional and commuter rail services report to the Federal Railroad Administration (FRA), rather than the Federal Transit Administration (FTA); therefore, data for these routes is not included in this report.

Trends

Figure C-5 shows the trends in transit asset conditions between 2019 to 2023. During this time, the percentage of the region's support vehicles in deficient condition varied between 36.0 and 47.9 percent, and the percentage of the region's Revenue Vehicles in deficient condition varied between 17.5 and 28.6 percent. Only marginal percentages of the region's passenger and administrative transit facilities and rail tracks were in deficient condition during that period. The condition of the region's transit facilities, vehicles, rail tracks, and other transit assets is critical to service reliability and appealing to potential riders. Worsening conditions can lead to less reliability, a loss of ridership, and strain the operating budgets of the region's transit agencies. This, in turn, limits the ability to improve service.

Figure C-5: Percent of Transit Assets in Deficient Condition, by Asset Category (All Providers)



Source: Federal Transit Administration's National Transit Database, 2019–2023.

DVRPC and Transit Agency Efforts Towards Transit Assets Targets

Transit providers are required to develop a TAMP to identify local funding prioritization. TAMP places value and understanding on the negative impacts of deferring maintenance and the positive outcomes of optimizing investment decisions that improve SGR. Successfully implementing TAMP requires using resources more efficiently to reduce an agency's environmental footprint, managing waste responsibly, building and supporting healthy places, and becoming more resilient to prepare for extreme weather.

MPOs are required to demonstrate how funding will be allocated to assist the transit agencies in meeting the TAMP targets. For a list of TIP-funded projects related to TAMP, see chapter four of the New Jersey and Pennsylvania TIPs.

Transit Safety

Targets and Reporting Requirements

Transit agencies are required to develop a Public Transit Agency Safety Plan (PTASP) that establishes performance targets for the five Safety Performance Targets (SPT) included in FTA's *National Transit Public Safety Plan* (NTPSP). Transit agencies are required to report their targets and performance to their respective state DOTs and MPOs to prioritize funding to improve

transit safety performance. FTA does not impose penalties for failing to meet SPTs set by transit providers.

Current targets and performance data can be found on DVRPC's TPM website under the [Transit Safety](#) tab.³² Transit agencies and states must identify SPTs by mode for each of the following categories:

Fatalities: Total number of fatalities reported to the NTD and rate per total vehicle revenue miles (VRM) by mode.

The transit safety performance measure requires that transit providers set annual targets for the number of fatalities that occur on each mode of transit that the agency operates, excluding deaths that result from trespassing, suicide, or natural causes. The National Public Transportation Safety Plan defines the modes as rail, fixed guideway bus service, and non-fixed route bus service. Fatalities are required to be calculated for both the total number of fatalities and the fatality rate per VRM. Transit worker fatalities are also reported, using the same metrics as total fatalities. SEPTA has chosen to include fatalities from trespassing and suicide in their reporting.

Injuries: Total number of injuries reported to NTD and rate per total VRM by mode.

The PTASP requires that transit agencies set annual targets for the number of injuries that occur on each mode of transit that the agency operates. Injuries are defined as "harm to a person

³² Delaware Valley Regional Planning Commission. *Transportation Performance Management: Transit Safety*. www.dvrpc.org/tpm/?indicator=ptasp.

that requires immediate medical attention away from the scene.” Injuries are required to be calculated for both the total number of injuries and the injury rate per VRM for each of the modes that the agency operates. Transit worker injuries are also reported, using the same metrics as total injuries.

Safety Events: Total number of safety events reported to NTD and rate per total VRM by mode.

Transit providers are required to set annual targets for the number and rate of safety events by mode that occur across the transit agency’s system. A safety event is defined by FTA as a “collision, derailment, fire, hazardous material spill, or evacuation.” Safety events are required to be calculated for both the total number of events and the event rate per VRM for each of the modes that the agency operates. Pedestrian, vehicle, and total collisions are also reported, using the same metrics as total safety events.

System Reliability: Mean distance between major mechanical failures by mode.

Transit providers are required to set annual targets for the agency’s system reliability for each mode of transit that the agency operates. The system reliability performance measure accounts for major mechanical failures that prevent the vehicle from starting or completing a scheduled trip. Mechanical failures and service interruptions can create hazardous conditions for transit operators and passengers, depending on the location of the interruption and whether passengers are required to alight in unsafe locations.

Assaults on Transit Workers: Total number of assaults on transit workers reported to the NTD and assault rate per VRM by mode.

An April 2024 update to the PTASP regulation changed safety performance target requirements focusing on transit worker safety. Transit providers are required to set annual targets for the total number of assaults on transit workers and the assault rate per vehicle revenue mile. Reporting requirements account for physical and non-physical assaults on any transit worker, including operators, police, or station agents, in any work setting.

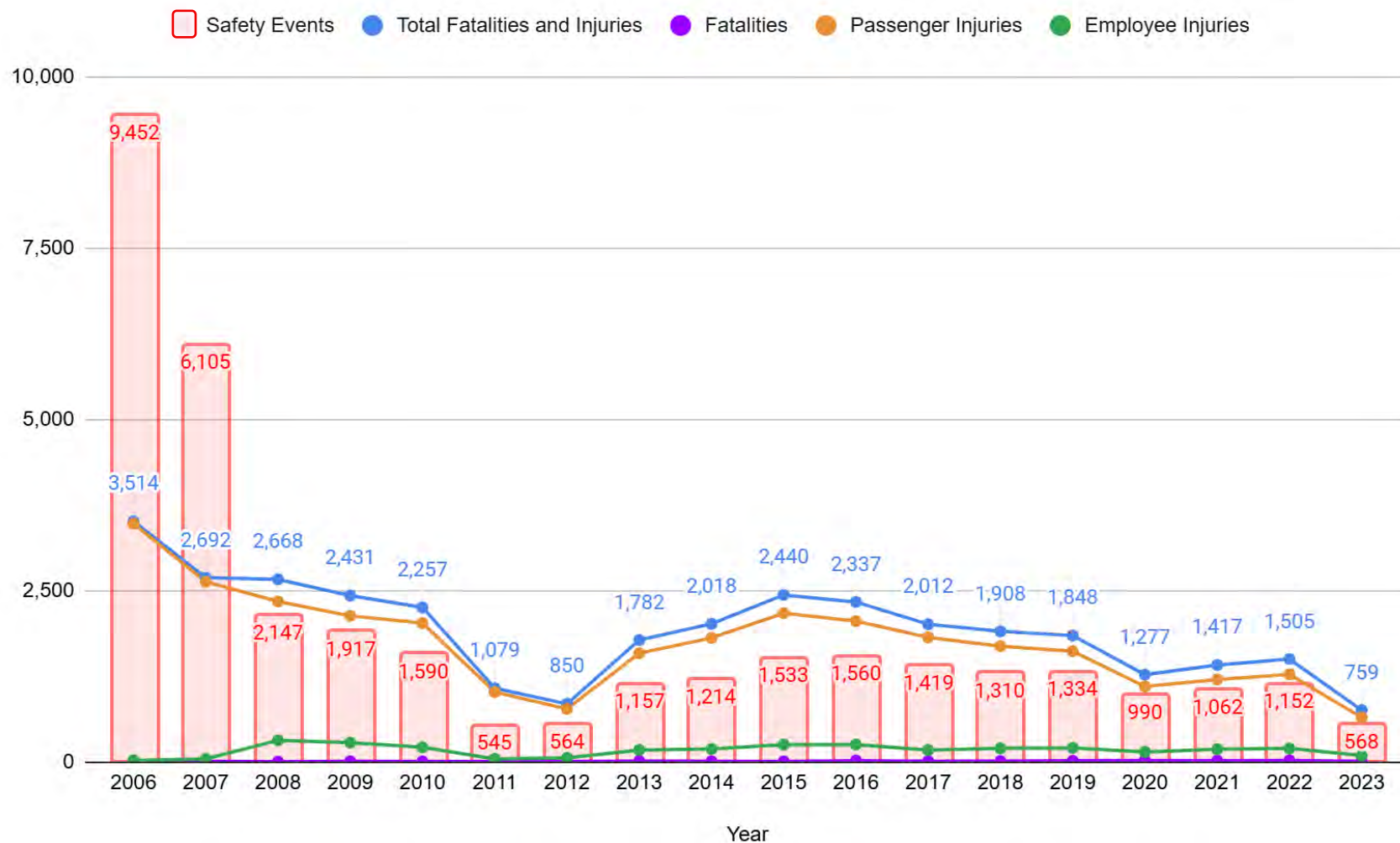
Trends

Safety across all transit modes has improved over the last two decades (See Figure C-6). Since 2018, fewer than 2,000 fatalities or injuries have been reported annually. Over 2,000 fatalities and injuries were reported in nine of the twelve years before 2018. Fatalities account for a small portion of the KSI figure (ranging between 9 and 24 yearly fatalities). Passenger injuries account for the majority (between 85 and 99 percent) of total injuries and fatalities. The lowest number of fatalities and injuries (759 total) was recorded in 2023.

Injuries to transit agency employees across all modes and agencies in the region totaled 4.3 per 200,000 employee-work hours in 2022. This rate peaked in 2018, then decreased to its lowest level in this dataset in 2020, during pandemic shutdowns. Regionwide, there has been a 15.7 percent decrease since 2009,

a 35.8 percent decrease since 2018, but a 79.2 percent increase since 2020.

Figure C-6: Safety Events and KSI on Transit in the DVRPC Region



Source: New Jersey Department of Transportation, Pennsylvania Department of Transportation, U.S. Census Bureau's Population Estimates Program, 2006–2023.

Similar trends were seen with the number of people killed and injured per 100,000 transit vehicle-revenue-miles. There was a notable peak of 20.5 KSI per 100,000 vehicle-revenue miles across all transit modes and agencies in the region in 2009, which then fell by roughly 80 percent to 4.2 in 2012. It then more than doubled to 9.6 in 2016 before nearly halving again to 4.5 in 2020 during the pandemic. Finally, the measure increased to 5.2 in 2023.

Safety Events have also decreased during this period. Total Safety Events follow similar trends to transit fatalities and injuries. Transit safety events peaked at 9,452 across all transit modes and agencies in the region in 2006, in the years included in this dataset. Since then, counts have been much lower, with an all-time low of 545 in 2011, a slight resurgence of 1,560 in 2016, and a return to 568 in 2023. This final count is a 95.9 percent decrease from the 2006 peak, and roughly a third of the 2016 count.

DVRPC and Transit Agency Efforts Towards Transit Safety Targets

DVRPC's goal to achieve Vision Zero—no transportation-related deaths or serious injuries—by 2050, encompasses transit incidents.

SEPTA has developed and implemented various safety programs, rules, and standard operating procedures. In addition to these administrative controls, SEPTA invests capital funds in various projects to maintain SEPTA's SGR and reduce risks,

improve safety, and help achieve safety performance target goals.

On the New Jersey side, NJ Transit and DRPA/PATCO are implementing projects that will help reduce rail vehicle collisions and improve passenger safety for all transit users in and around NJ Transit and DRPA/PATCO's operating environments. NJ Transit takes every precaution to ensure both passenger and public safety on its bus, rail, and light rail systems. NJ Transit operates a risk-based safety management system (SMS), a data-driven process that proactively manages risks in the public transportation system. The SMS aims to transform the safety culture by making safety everyone's responsibility, empowering employees to take an active role in safety, and encouraging employees and contractors to report safety concerns to senior management.

For a list of TIP-funded projects related to Transit Safety, see chapter four of the New Jersey and Pennsylvania TIPs.